sPHENIX Self-Paced Course

Module 1: The sPHENIX physics program: QGP, jets, heavy flavour, cold QCD and spin

Prepared for P. D. Lewis (Ohio University) November 2026

Self-contained study text based on the Summer-2026 sPHENIX School (BNL, July 2026), public sPHENIX papers and standard textbook physics

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How to use this module

This module is self-contained. You do not need the school slides, recordings or papers to follow it: every concept is built up here from first principles to sPHENIX practice, with the reasoning, short derivations and worked numbers. The school talks listed below are the *sources* that the text paraphrases; they are cited as [S##] so that you can trace a statement back if you ever want to. Read the module straight through once. Then reread it and answer the self-test questions (Section 11) before looking at the answers in Appendix A. The hands-on exercises are optional; nothing in the text depends on doing them.

Learning objectives

After this module you should be able to:

O1. State what the quark–gluon plasma (QGP) is, why it exists in QCD, and why RHIC and the LHC are complementary.

O2. Use the basic kinematic variables (p_{T}, y, η, $\sqrt{s_{NN}}$, invariant cross section) fluently, with numbers.

O3. Explain the Bjorken energy density, collective flow (v_n, eccentricity, cumulants) and what the Bulk topical group measures.

O4. Derive the key scalings of jet quenching (broadening $\hat{q}L$, formation time, $\omega_c = \hat{q}L^2$, $\Delta E \propto L^2$, colour coherence) and connect them to R_{AA}, dijets and substructure.

O5. Explain why heavy quarks and quarkonia probe the QGP, and what the Heavy Flavor topical group needs from tracking.

O6. Explain transverse single-spin asymmetries, the Sivers and Collins mechanisms, twist-3 correlators, and the cold-QCD channels in sPHENIX.

O7. Summarise the history from PHENIX to sPHENIX and the organisation into topical groups and paper preparation groups.

O8. Place neutral-meson measurements (π^0, η cross sections, R_{AA}/R_{CP}, A_N, v_2) in the whole programme.

Prerequisites, time and sources

Prerequisites: none within the course (this is Module 1). Undergraduate special relativity and quantum mechanics. **Estimated time:** 9–10 hours (reading 6 h, self-test 2 h, optional exercises 1–2 h).

School talks used as sources (all "sPHENIX School 2026", BNL, July–August 2026):

- [S01] J. Huang, M. Connors, "Welcome to sPHENIX school and sPHENIX Overview".
- [S02] D. Morrison, "sPHENIX History".
- [S03] Y. Mehtar-Tani, "sPHENIX Theory Overview: exploring QGP dynamics with jets".
- [S04] G. Nukazuka, M. Sarsour, "Cold QCD TG Analysis Opportunities".
- [S05] M. Mitrankova, R. Belmont, "Bulk TG Analysis Opportunities".
- [S06] A. Frawley, C. Dean, G. Ottino, M. Rosati, "Heavy Flavor TG Analysis Opportunities".
- [S07] J. Putschke, Y. Go, "Jet Structure TG Analysis Opportunities".
- [S44] G. Nukazuka, "Cold QCD Lecture Series announcement"; [S50] M. Cerutti, "Cold-QCD lecture (workfest)".
- Supporting: [S08] J. Haggerty (sPHENIX timeline), [S48] K. Hock (polarized protons at RHIC).

Public readings (optional): the first two sPHENIX Au+Au papers on charged-hadron multiplicity [1] and transverse energy [2]; the first p+p papers on dijet imbalance [3] and isolated photons [4]; the sPHENIX proposal [5]; overview talks [6, 7].

1 Why sPHENIX exists: QCD, the QGP and hard probes

Goal. Understand the physics question that sPHENIX was built to answer, starting from the basic properties of QCD.

1.1 QCD in one page

Quantum chromodynamics (QCD) is the theory of quarks and gluons. Quarks carry one of three "colour" charges; gluons carry colour themselves and therefore interact with each other. That self-interaction is the origin of the two properties that dominate everything in this course.

Asymptotic freedom. The strong coupling decreases with the momentum transfer Q of a process. At one loop

$$\alpha_s(Q^2) = \frac{12\pi}{(33 - 2n_f)\,\ln(Q^2/\Lambda^2_{\rm QCD})}, \qquad \Lambda_{\rm QCD} \approx 0.2\ {\rm GeV}, \tag{1}$$

with n_f active quark flavours. For $Q = 10$ GeV and $n_f = 5$ this gives $\alpha_s \approx 12\pi/(23\ln 2500) \approx 0.21$; for $Q = 1$ GeV it gives ≈ 0.5. Hard processes (large Q) are therefore calculable in perturbation theory (pQCD).

Confinement. At low Q (large distances) the coupling is large; isolated colour charges are never observed. Quarks and gluons appear only inside hadrons. The energy stored in the colour field between separating charges grows roughly linearly with distance, until it is cheaper to create a new quark–antiquark pair; this is *hadronization*.

Deconfinement. Lattice QCD (numerical QCD on a space-time grid) shows that at temperatures above $T_c \approx 155$ MeV (about 2×10^{12} K) hadronic matter turns into a medium where colour is screened over distances of the order of the inter-particle spacing. This medium is the quark–gluon plasma. The transition at zero net baryon density is a smooth crossover, not a sharp phase transition. The universe passed through this state a few microseconds after the Big Bang. The critical energy density is of order 1 GeV/fm^3, compared with about 0.15 GeV/fm^3 in ordinary nuclear matter and about 0.5 GeV/fm^3 inside a proton [S05].

1.2 What RHIC learned, and the open question

RHIC began colliding nuclei in 2000 [S02]. Two discoveries defined the field. First, the produced matter flows collectively, and relativistic viscous hydrodynamics describes the flow only if the shear viscosity to entropy density ratio is very small, $\eta/s \approx 0.1$, close to the conjectured quantum lower bound $1/4\pi \approx 0.08$. The QGP behaves as an almost perfect liquid, far less viscous (in this dimensionless sense) than water or helium [S03]. Second, high-$p_{\rm T}$ hadrons are strongly suppressed in central Au+Au collisions relative to scaled p+p collisions; PHENIX first reported this with charged hadrons and π^0s [9, 10]. This is *jet quenching*.

These discoveries raise a deeper question. A perfect liquid is a strongly coupled system with no well-defined quasi-particles; but at very short distances asymptotic freedom says the QGP must look like weakly coupled quarks and gluons. *How does the liquid-like behaviour emerge from QCD as a function of the resolution scale?* [S03] Answering this needs probes that resolve the medium at different length scales: jets of different energies and radii, heavy quarks of different masses, and quarkonium states of different sizes.

1.3 The sPHENIX answer

The 2015 US Nuclear Physics Long Range Plan stated that a central goal of RHIC and the LHC is to probe the QGP at shorter and shorter length scales, and that a state-of-the-art jet detector at RHIC—sPHENIX—is essential for that; the 2023 plan asked to complete the sPHENIX programme as part of concluding the RHIC science mission [S01]. The design follows directly from the physics [S01]:

- **Central calorimetry** (EMCal + inner and outer HCal, $|\eta| < 1.1$, full azimuth) for full jet reconstruction and for photons and electrons.
- **Central tracking** (MVTX, INTT, TPC, TPOT) in a 1.4 T solenoid for heavy-flavour tagging, quarkonia and jet substructure with charged particles.
- **Forward detectors** (MBD, sEPD, ZDC) for triggering and event characterisation (vertex, centrality, event plane).
- **A high-rate DAQ**, including streaming readout of the trackers, to record very large minimum-bias samples.

Why RHIC when the LHC exists?

The LHC reaches much higher energy, so hard probes are more abundant there. RHIC is complementary for three reasons. (1) The QGP at RHIC is produced closer to T_c, where the medium is expected to be most strongly coupled; comparing RHIC and LHC tests the temperature dependence of transport properties. (2) At RHIC the underlying event is smaller and jets of a given p_{T} come from a different mix of quarks and gluons and a different region of the parton distributions. (3) RHIC uniquely provides polarized proton beams, so the same detector can run a spin programme. sPHENIX brings LHC-style calorimetric jet reconstruction to RHIC energies [S01,S03].

Checks.

You understand this section if you can explain, without notes: (a) why pQCD works for jets but not for the bulk; (b) what "almost perfect liquid" means in terms of η/s; (c) why a short-distance probe is needed to answer the emergence question.

2 Kinematics toolkit

Goal. Be fluent with the variables every sPHENIX analysis uses, with numbers you can check in your head.

2.1 Variables

Take the beam along z. For a particle with energy E and momentum $\vec{p}$:

$$p_{\mathrm{T}} = \sqrt{p_x^2 + p_y^2}, \qquad m_{\mathrm{T}} = \sqrt{p_{\mathrm{T}}^2 + m^2}, \tag{2}$$

$$y = \tfrac{1}{2} \ln \frac{E + p_z}{E - p_z}, \qquad \eta = -\ln \tan(\theta/2) = \tfrac{1}{2} \ln \frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}. \tag{3}$$

Rapidity y is additive under boosts along z: a boost changes every y by the same constant, so shapes of dN/dy are boost invariant. Pseudorapidity η depends only on the polar angle, which is what a detector measures; $y \to \eta$ when $p_{\mathrm{T}} \gg m$. Useful relations: $E = m_{\mathrm{T}} \cosh y$, $p_z = m_{\mathrm{T}} \sinh y$.

Worked numbers. The central barrel covers $|\eta| < 1.1$, i.e. polar angles $36.8^\circ < \theta < 143.2^\circ$. The sEPD covers $2.0 < |\eta| < 4.9$, i.e. 15.4° down to 0.85° from the beam. The MBD covers about $3.5 < |\eta| < 4.6$,

i.e. 3.4° to 1.1° [S08]. A 100 GeV/nucleon Au beam has $y_{\rm beam} = \cosh^{-1}(E/m_N) \approx \ln(\sqrt{s_{NN}}/m_N) = \ln(200/0.938) \approx 5.36$.

2.2 Centre-of-mass energy and the invariant cross section

For symmetric collisions $\sqrt{s} = 2E_{\rm beam}$. For nuclei one quotes the energy per nucleon pair, $\sqrt{s_{NN}}$; RHIC Au+Au, O+O and p+p in sPHENIX were all at $\sqrt{s_{NN}} = 200$ GeV. The Lorentz-invariant differential cross section is

$$E\frac{d^3\sigma}{dp^3} = \frac{1}{2\pi p_{\rm T}}\frac{d^2\sigma}{dp_{\rm T}\,dy}. \tag{4}$$

The factor $1/(2\pi p_{\rm T})$ comes from $d^3p = p_{\rm T}\,dp_{\rm T}\,d\phi\,dp_z$ and $dp_z/E = dy$, integrated over ϕ. In an experiment, with N counts in a bin, integrated luminosity $\mathcal{L}$, efficiency ε, acceptance A and branching ratio B,

$$E\frac{d^3\sigma}{dp^3} = \frac{1}{2\pi p_{\rm T}}\,\frac{N}{\mathcal{L}\,\varepsilon A\,B\,\Delta p_{\rm T}\,\Delta y}. \tag{5}$$

For heavy ions the per-event invariant yield replaces $\sigma/\mathcal{L}$ by $N/N_{\rm evt}$.

2.3 Scaling variable x_T

In a $2 \to 2$ hard scattering at $y = 0$ a hadron of transverse momentum $p_{\rm T}$ probes parton momentum fractions of order $x_T = 2p_{\rm T}/\sqrt{s}$. A 10 GeV π^0 at RHIC has $x_T = 0.1$; the same x_T at the LHC (5.02 TeV) corresponds to $p_{\rm T} \approx 250$ GeV. This is one reason RHIC probes a different quark/gluon mix: at fixed $p_{\rm T}$, RHIC is at larger x, where quarks matter more.

Connect to your analysis.
Your p+p π^0/η cross section is exactly Eq. (4) with $B = B(\pi^0 \to \gamma\gamma)$ or $B(\eta \to \gamma\gamma)$, $\mathcal{L}$ from the GL1 scalers, εA from PYTHIA8 with EMCal smearing, and the trigger efficiency folded in. The $|y|$ window of the pair (not $|\eta|$ of each photon) defines Δy. Always check that your εA is defined with the same Δy and the same $p_{\rm T}$ binning as the raw yield.

3 The life of a heavy-ion collision and the Bulk programme

Goal. Know the stages of a heavy-ion collision, how its global properties (energy density, multiplicity, flow) are measured, and which measurements the Bulk topical group (TG) pursues in sPHENIX.

3.1 Stages of the collision

Figure 1 is the "standard model" of a heavy-ion collision [S05]. Two Lorentz-contracted nuclei (at 100 GeV/nucleon the contraction factor is $\gamma \approx 107$, so a gold nucleus of radius 7 fm is about 0.13 fm thick) pass through each other. Hard scatterings (jets, heavy quarks, prompt photons) happen first, within $\sim 1/p_{\rm T}$. A dense, initially far-from-equilibrium gluon system (often called the glasma) forms. Within about 1 fm/c it becomes well described by hydrodynamics; modern theory calls this *hydrodynamization* rather than thermalization, because hydrodynamics can work before full local equilibrium [S05]. The QGP expands and cools for a few fm/c, hadronizes near T_c, and the hadrons stop changing species (chemical freeze-out) and then stop interacting (kinetic freeze-out). The detector sees only the final hadrons, leptons and photons.

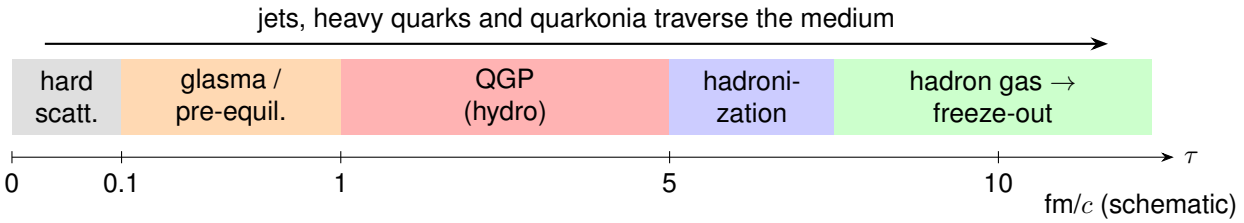


Figure 1: Schematic time line of a central heavy-ion collision at RHIC (not to scale; times are order-of-magnitude).

3.2 Bjorken picture and energy density

Two pictures of stopping compete [S05]. In Landau's picture the nuclei stop each other and dN/dy is roughly Gaussian. In Bjorken's picture the nuclei pass through each other and leave a boost-invariant region of produced matter; dN/dy is flat near $y = 0$ [11]. RHIC data (PHOBOS, BRAHMS) show that the Bjorken picture is the right starting point at 200 GeV [S05].

Derivation. In the boost-invariant picture, the matter at rapidity y is at $z = t \tanh y$, i.e. at proper time $\tau = \sqrt{t^2 - z^2}$ a slice dy occupies a longitudinal length $dz = \tau\, dy$ (at $y \approx 0$). If the transverse overlap area is A_T, the volume of the slice is $A_T \tau\, dy$ and the energy in it is dE_T. Hence

$$\varepsilon_{\mathrm{Bj}}(\tau) = \frac{1}{\tau A_T}\frac{dE_T}{dy}, \qquad E_T = \sum_i E_i \sin\theta_i. \tag{6}$$

Worked example (round numbers). For a central Au+Au collision take $A_T = \pi R^2$ with $R = 1.18 A^{1/3}$ fm $= 6.9$ fm, so $A_T \approx 148$ fm^2. With $dE_T/d\eta \approx 600$ GeV and a Jacobian $dE_T/dy \approx 1.25\, dE_T/d\eta$, at $\tau = 1$ fm/c one gets $\varepsilon \approx (1.25 \times 600)/148 \approx 5$ GeV/fm^3, several times the critical value. These are illustrative round numbers of the size found by the RHIC experiments. Because A_T is fairly well determined from centrality (Glauber modelling, Module 2) but τ is model dependent, experiments often quote the product $\varepsilon\tau$ [S05].

3.3 How sPHENIX measured $dE_T/d\eta$ and $dN_{ch}/d\eta$

The first two sPHENIX physics papers came from the Bulk TG [S05]:

- **Transverse energy** [2]. Each calorimeter layer (EMCal, inner HCal, outer HCal) measures $\sum E_{T,\mathrm{tower}}(\eta)$. A correction factor $C(\eta) = \sum E_{T,\mathrm{tower}} / \sum E_{T,\mathrm{particle}}$, built from HIJING events reweighted to match measured identified-particle spectra, converts the tower sum into the true $dE_T/d\eta$. The EMCal sees the largest fraction of the total energy, the outer HCal much less, and the inner HCal (mostly aluminium) only a small fraction. Agreement between the EMCal-based and HCal-based results is a strong internal check.
- **Charged-hadron multiplicity** [1]. Using Run-2024 Au+Au data taken with the magnet off (straight tracks), charged hadrons were counted as pairs of clusters in the inner and outer INTT layers. $dN_{ch}/d\eta$ was obtained with two very different methods with different systematics; their agreement shows systematic control. The result agrees with earlier PHOBOS data, and the centrality dependence is not a simple scaling with the number of participants.

Two-component model. A common parametrization is $dN_{ch}/d\eta = n_{pp}\left[(1-x)\,\frac{N_{\mathrm{part}}}{2} + x\, N_{\mathrm{coll}}\right]$, where N_{part} (participants) and N_{coll} (binary collisions) come from Glauber modelling and x is the "hard" fraction. As a toy check, with $n_{pp} = 2.4$, $x = 0.13$, $N_{\mathrm{part}} = 350$ and $N_{\mathrm{coll}} = 1050$ one gets about 690. The model is not the end of the story; participant-quark scaling is an alternative and the question is not settled [S05].

3.4 Collective flow

The overlap region of a non-central collision is almond shaped. Pressure gradients are larger along the short axis, so more particles are pushed in that direction. Expanding the azimuthal distribution in a Fourier series around the symmetry planes Ψ_n,

$$\frac{dN}{d\phi} \propto 1 + \sum_{n=1}^{\infty} 2v_n \cos[n(\phi - \Psi_n)], \qquad v_n = \langle \cos[n(\phi - \Psi_n)] \rangle. \tag{7}$$

The initial shape is quantified by the eccentricities $\varepsilon_n = \sqrt{\langle r^n \cos n\phi \rangle^2 + \langle r^n \sin n\phi \rangle^2} / \langle r^n \rangle$, averaged over the initial energy (or nucleon) density [S05]. Hydrodynamics converts shape into momentum anisotropy; to a good approximation $v_2 \approx \kappa_2 \varepsilon_2$ and $v_3 \approx \kappa_3 \varepsilon_3$, where the response coefficients κ_n decrease with viscosity and with n (viscosity damps fine structure first). This is how flow constrains η/s.

Centrality dependence of v_2. v_2 rises from central to mid-central collisions and falls again in peripheral ones. Two effects compete: more peripheral means larger ε_2 (higher v_2), but also a smaller system with fewer interactions and less time to build flow (lower v_2) [S05].

Fluctuations and odd harmonics. Nuclei are made of nucleons at random positions, so the initial state is lumpy. Event-by-event fluctuations give non-zero odd eccentricities (ε_3, triangular flow v_3) and symmetry planes Ψ_n that differ between harmonics [12]. Measured v_3 is indeed non-zero [S05].

Cumulants. Multi-particle correlations measure moments of the v_n distribution and suppress non-flow (few-particle correlations such as jets and decays). If v_2 fluctuates around $\bar{v}$ with small variance σ^2,

$$v_2\{2\}^2 = \langle v_2^2 \rangle = \bar{v}^2 + \sigma^2, \tag{8}$$

$$v_2\{4\}^4 = 2\langle v_2^2 \rangle^2 - \langle v_2^4 \rangle \approx (\bar{v}^2 - \sigma^2)^2, \tag{9}$$

so $v_2\{2\} \approx \sqrt{\bar{v}^2 + \sigma^2}$ and $v_2\{4\} \approx v_2\{6\} \approx v_2\{8\} \approx \sqrt{\bar{v}^2 - \sigma^2}$. Small deviations between $v_2\{4\}, v_2\{6\}, v_2\{8\}$ measure the non-Gaussian shape of the fluctuations; the ratio $\sigma_{v_2}/\langle v_2 \rangle$ tests initial-state models. In very central collisions the small-variance approximation breaks down; in peripheral ones the hydro response becomes non-linear [S05]. With minimum-bias Au+Au samples far larger than those available to PHENIX, sPHENIX can push these measurements to high precision and to higher harmonics and correlations between harmonics (normalized symmetric cumulants) [S05].

3.5 Small systems and the ridge

In two-particle correlations versus $\Delta\eta$ and $\Delta\phi$, a near-side structure extended over many units of $\Delta\eta$ (the *ridge*) is a signature of collective flow; it was seen first in Au+Au at RHIC and later in high-multiplicity p+p at the LHC [S05]. Whether a ridge exists in p+p at RHIC is an open question that the large sPHENIX minimum-bias and high-multiplicity samples can address. Collision geometry differs between small systems: d+Au has a large intrinsic ε_2 from the two-nucleon projectile, while O+O acquires ε_3 only through fluctuations. The sPHENIX O+O data are therefore valuable for separating geometry from response [S05].

3.6 How it is done in sPHENIX

- Acceptance: the central barrel ($|\eta| < 1.1$) for particles of interest, and the sEPD ($2.0 < |\eta| < 4.9$) or MBD for symmetry planes. A large rapidity gap between the two suppresses non-flow [S05].
- Event-plane method: Q-vectors $Q_n = \sum_j w_j e^{in\phi_j}$ from sEPD tiles give Ψ_n; the observed $v_n^{\rm obs} = \langle \cos n(\phi - \Psi_n) \rangle$ is divided by the event-plane resolution (Module 2). In coresoftware the module is `EventPlaneReco` (package `eventplaneinfo`); centrality comes from `CentralityReco`.

- Neutral mesons: π^0 v_2 uses the same EMCal diphoton pairs as your R_{AA} analysis, binned in $\phi - \Psi_2$; the signal yield per bin is fitted and the modulation extracted. It is an EMCal performance benchmark against PHENIX measurements [S05].
- Multiplicity in O+O: ongoing analyses use silicon tracking (see Module 10 case studies [S25,S26]).

Checks.
(1) A flow analysis must show that v_n does not depend on the choice of event-plane detector once resolution-corrected (north vs south sEPD, MBD vs sEPD). (2) $v_2\{4\} < v_2\{2\}$ is expected; the opposite usually signals a bug or non-flow. (3) $dE_T/d\eta$ from the EMCal and from the HCal must agree within systematic uncertainties.

Pitfalls.
Confusing η and y when converting $dE_T/d\eta$ to dE_T/dy (the Jacobian depends on the particle mix). Forgetting that v_n from a single event plane mixes $\bar{v}$ and fluctuations differently from cumulants. Ignoring non-flow in peripheral and small systems.

4 Jets as probes of the QGP: theory of jet quenching

Goal. Understand, with simple derivations, why and how energetic partons lose energy in the QGP, and what each class of jet observable is sensitive to. This is the core physics motivation of sPHENIX [S03].

4.1 Jets in vacuum: a calibrated probe

A jet is the collimated spray of hadrons produced when an energetic quark or gluon showers and hadronizes. After five decades of work—from the discovery of the gluon in three-jet events at DESY in 1979 to the Higgs discovery—jet cross sections and shapes in vacuum are predicted with few-percent precision [S03]. In p+p collisions the cross section factorizes:

$$d\sigma^{pp\to hX} = \sum_{a,b,c} f_a(x_1,\mu^2) \otimes f_b(x_2,\mu^2) \otimes d\hat{\sigma}_{ab\to cX}(\mu^2) \otimes D_{c\to h}(z,\mu^2), \tag{10}$$

with parton distribution functions (PDFs) f, a perturbative partonic cross section $\hat{\sigma}$, and a fragmentation function D giving the probability that parton c produces hadron h with momentum fraction z. For a reconstructed jet the fragmentation function is replaced by the jet algorithm acting on the shower. Because the vacuum reference is so well known, any modification in a nucleus–nucleus collision can be attributed to the medium. The analogy is Rutherford's experiment: a beam of well-understood probes reveals the inner structure of the target [S03].

4.2 The nuclear modification factor

$$R_{AA}(p_{\mathrm{T}}) = \frac{1}{\langle T_{AA}\rangle}\frac{dN^{AA}/dp_{\mathrm{T}}}{d\sigma^{pp}/dp_{\mathrm{T}}} = \frac{dN^{AA}/dp_{\mathrm{T}}}{\langle N_{\mathrm{coll}}\rangle\, dN^{pp}/dp_{\mathrm{T}}}, \qquad \langle T_{AA}\rangle = \frac{\langle N_{\mathrm{coll}}\rangle}{\sigma_{\mathrm{inel}}^{NN}}. \tag{11}$$

$R_{AA} = 1$ means the nucleus–nucleus collision is an incoherent superposition of nucleon–nucleon collisions. Prompt photons, which do not interact strongly, should give $R_{AA} \approx 1$ and test the N_{coll} scaling. When no p+p reference is available, the central-to-peripheral ratio $R_{CP} = (\langle N_{\mathrm{coll}}^{\mathrm{per}}\rangle\, dN^{\mathrm{cen}})/(\langle N_{\mathrm{coll}}^{\mathrm{cen}}\rangle\, dN^{\mathrm{per}})$ is used.

Why a modest energy loss gives a large suppression. Suppose the p+p spectrum is a power law $dN/dp_{\rm T} \propto p_{\rm T}^{-n}$ and every parton loses a constant fraction ϵ of its momentum. A hadron observed at $p_{\rm T}$ came from $p_{\rm T}/(1-\epsilon)$, so

$$R_{AA} = \frac{(p_{\rm T}/(1-\epsilon))^{-n}}{p_{\rm T}^{-n}} \frac{d(p_{\rm T}/(1-\epsilon))}{dp_{\rm T}} = (1-\epsilon)^{n-1} \tag{12}$$

for the yield per unit $p_{\rm T}$. If instead the invariant yield falls as $E\,d^3N/dp^3 \propto p_{\rm T}^{-n}$, then $dN/dp_{\rm T} \propto p_{\rm T}^{-(n-1)}$ and $R_{AA} = (1-\epsilon)^{n-2}$. Toy numbers: with $n = 8$ and $\epsilon = 0.2$, $(0.8)^6 \approx 0.26$. A 20% energy loss produces a factor four suppression, because the spectrum is steep. This also shows why R_{AA} alone cannot separate "how much" from "how often" energy is lost.

4.3 Microscopic mechanisms

Elastic energy loss. Bjorken predicted jet quenching in 1982 as a consequence of elastic collisions of the energetic parton with plasma constituents [S03]. Collisional loss is important for heavy quarks and at moderate energy.

Transverse momentum broadening. The medium screens colour over the Debye length $\xi_D \sim 1/(gT)$ and the parton scatters with mean free path $\ell_{\rm mfp} \sim 1/(g^2T)$. Each scattering gives a small transverse kick of size $\sim m_D$. After $L/\ell_{\rm mfp}$ independent kicks (a random walk, Brownian motion in transverse momentum) [S03]

$$\langle k_\perp^2 \rangle \simeq \hat{q}\,L, \qquad \hat{q} \sim \frac{m_D^2}{\ell_{\rm mfp}}, \tag{13}$$

which defines the *jet quenching transport coefficient* $\hat{q}$: the mean transverse momentum squared acquired per unit length.

Medium-induced radiation and the LPM effect. Multiple scattering also induces gluon radiation. A gluon of energy ω and transverse momentum $k_\perp$ needs a *formation time* $t_f \simeq \omega/k_\perp^2$ to separate from its parent (uncertainty principle). During formation the gluon itself is broadened: $k_\perp^2 \simeq \hat{q}\,t_f$. Solving the two relations self-consistently,

$$t_f \simeq \sqrt{\omega/\hat{q}}. \tag{14}$$

If $t_f \gg \ell_{\rm mfp}$, many scatterings act coherently as a single effective one; independent emissions are suppressed by destructive interference. This is the Landau–Pomeranchuk–Migdal (LPM) effect, worked out for QCD by Baier–Dokshitzer–Mueller–Peigné–Schiff and Zakharov (BDMPS-Z) and others [S03]. Emission is possible only if the gluon forms inside the medium, $t_f < L$, which gives a maximum (characteristic) energy

$$\omega_c = \hat{q}\,L^2. \tag{15}$$

The spectrum per unit energy behaves as $\omega\,dI/d\omega \simeq (\alpha_s C_R/\pi)\sqrt{\omega_c/\omega}$ for $\omega < \omega_c$ ($C_R = 4/3$ for quarks, 3 for gluons). Integrating $\int^{\omega_c} d\omega\,\sqrt{\omega_c/\omega} \propto \omega_c$ gives

$$\Delta E \simeq \alpha_s C_R\,\hat{q}\,L^2, \tag{16}$$

the famous L^2 dependence of radiative loss in a static medium (in an expanding medium the dependence is weaker). The C_R factor means gluons lose about $9/4$ times more energy than quarks.

Colour coherence. A jet is not a single colour charge: after its first splitting it consists of several charges separated by an angle θ. The medium resolves them only if their transverse separation $r_\perp = \theta L$ exceeds the medium resolution length $\Lambda_{\rm med} \sim (\hat{q}L)^{-1/2}$ [S03]. This gives a critical angle

$$\theta_c \simeq \frac{1}{\sqrt{\hat{q}L^3}}. \tag{17}$$

Splittings with $\theta < \theta_c$ are unresolved: the pair loses energy as one effective charge (coherence). Splittings with $\theta > \theta_c$ are resolved: each prong radiates independently and the jet loses more energy (decoherence). The medium acts like a microscope that counts the effective number of colour charges in a jet [S03].

From jet energy to thermal energy. Medium-induced emissions cascade to softer and softer gluons with a constant energy flux, like a turbulent cascade, until the fragments reach the medium temperature and thermalize. Low-energy jets thermalize quickly and are strongly modified; high-energy jets are less modified [S03]. The energy is not destroyed: it reappears at large angles as soft particles (*medium response*).

Anomalous diffusion. Quantum corrections make $\hat{q}$ depend logarithmically on the transverse momentum scale, so broadening grows faster than L (super-diffusion). One consequence is a non-trivial dependence of jet quenching on system size, which makes comparisons of O+O and Au+Au at the same energy especially interesting [S03].

4.4 Phenomenology: what the data say so far

Two classes of observables exist [S03]. *Inclusive* observables: the single-inclusive jet spectrum and R_{AA}, dijets and γ+jet (the photon tags the initial parton energy). *Substructure* observables: jet mass, angularities, groomed momentum sharing and angle, energy correlators. The data from RHIC and the LHC suggest a remarkably simple picture [S03]:

1. strong energy loss (jets and hadrons are suppressed by large factors);
2. weak dependence of jet R_{AA} on the cone size R, which is sensitive to colour coherence and to how lost energy is recovered at larger R;
3. modest modification of substructure;
4. jets that survive look more collimated—either colour decoherence, or a selection bias (gluon jets lose more energy and drop out of the sample, leaving narrower quark jets);
5. a measurable medium response.

4.5 Substructure observables in brief

For a two-prong jet with momentum fractions z and $1-z$ at opening angle θ, the jet mass is $m^2 \simeq z(1-z)p_{\mathrm{T}}^2\theta^2$. Grooming (soft drop) removes soft wide-angle radiation, which is sensitive to the underlying event: the clustering history is undone and soft branches are dropped until $\min(p_{\mathrm{T}}^1, p_{\mathrm{T}}^2)/(p_{\mathrm{T}}^1 + p_{\mathrm{T}}^2) > z_{\mathrm{cut}}(\Delta R/R)^\beta$; the surviving splitting defines z_g and θ_g. The energy–energy correlator (EEC) weights every pair of particles in a jet by their energies, $d\Sigma/d\theta = \sum_{i\neq j} \frac{p_{T,i}p_{T,j}}{p_{T,\mathrm{jet}}^2}\delta(\theta - \theta_{ij})$; it is less sensitive to background and cleanly shows the transition from perturbative showering (small θ scaling region) to hadronization [S03,S07]. The Lund plane maps every splitting in the plane of $\ln(1/\theta)$ and $\ln(k_T)$, separating hard/soft and collinear/wide-angle regions and giving access to formation times [S07].

4.6 Where theory is heading

Current phenomenology still relies heavily on Monte Carlo models. The goals are systematic effective theories of jet quenching with controlled higher-order corrections, a factorization that separates hard production (time $\sim 1/p_{\mathrm{T}}$) from medium-induced loss (time $\sim L$) [S03],

$$\frac{d\sigma^{AA}}{dp_{\mathrm{T}}} \simeq \int d\epsilon\, P(\epsilon)\, \frac{d\sigma^{pp}}{dp_{\mathrm{T}}}(p_{\mathrm{T}} + \epsilon), \tag{18}$$

where $P(\epsilon)$ is the probability to lose energy ϵ, and global Bayesian analyses that fit many observables together to extract $\hat{q}$ and its scale dependence, the system-size scaling of energy loss, thermalization and colour flow [S03].

Connect to your analysis.
Your Au+Au π^0 R_{CP} is a single-hadron inclusive observable. The toy model above shows why it is sensitive to the spectral slope: if your p_{T} reach changes the local slope n, the same energy loss gives a different R_{CP}. A π^0 comes mostly from the leading fragment of a jet ($z \sim 0.5$–0.7 at RHIC), so it probes the parton energy loss more directly than the full jet, but it cannot see where the lost energy goes. Your p+p cross section is the reference that turns R_{CP} into R_{AA}.

5 The Jet Structure topical group

Goal. Know what the Jet Structure TG (JSTG) measures in p+p, Au+Au and O+O, and where your analysis sits.

5.1 Organisation

sPHENIX physics is organised in four topical groups: Bulk, Jet Structure, Heavy Flavor and Cold QCD [S01]. A specific paper is produced by a *paper preparation group* (PPG), which goes through internal review (an internal review committee) and approval before preliminary results and papers are released. The JSTG oversees a large number of PPGs, including the first two sPHENIX p+p papers: dijet imbalance and acoplanarity [3] and isolated prompt photons [4] [S07].

5.2 The datasets

sPHENIX took data for three years, ending with the last RHIC run in February 2026 [S07,S08]: 2023 was commissioning with Au+Au; 2024 delivered p+p (a large calorimeter-only sample plus an all-subsystem sample) and Au+Au; 2025 delivered p+p and O+O. All were at $\sqrt{s_{NN}} = 200$ GeV. Run details are in Module 2.

5.3 Opportunities in p+p

- **Calorimeter-only jets and photons**: inclusive jets, dijets, γ+jet, the underlying event, and spin asymmetries of jets and photons (with Cold QCD).
- **Neutral-meson cross sections** (π^0, η): validate EMCal performance, test pQCD and constrain parton fragmentation, and provide the reference for neutral-meson suppression in heavy-ion collisions [S07]. This is your PhD analysis.
- **Substructure**: jet mass, angularities, groomed z_g and R_g, jet shape and girth, jet-axis differences (soft radiation vs hard splitting), Lund plane, EEC, track functions (the charged-momentum fraction of a jet, sensitive to correlated hadronization). Calorimeter-only substructure is limited by tower granularity; the move towards jets with tracks will improve angular resolution [S07].
- **Collectivity in high-multiplicity jets**: can collective behaviour arise inside a single fragmenting parton? [S07]

5.4 Opportunities in Au+Au and O+O

- **R-dependent jet R_{AA}**: LHC measurements show tension on how R_{AA} changes with R; RHIC's lower background and different quark/gluon mix can help [S07].
- **Dijet imbalance in O+O**: the LHC sees quenching features in O+O relative to p+p; the sPHENIX O+O sample is larger [S07].
- **Jet–hadron and γ–hadron correlations**: look for the medium response, including the "diffusion wake" (a depletion of soft particles opposite to the jet direction predicted by models) [S07].
- **Heavy-quark-tagged jets**, **jet v_n**, **γ+jet in Au+Au**.
- **Pre-equilibrium probes**: jets may already be quenched in the glasma; the width of jet-like di-hadron correlations and jet deformation by flow are candidate observables [S07].

Checks.
For any jet measurement ask: what is the jet definition (anti-k_t, R, constituents: towers, clusters, tracks)? How is the underlying event subtracted? How are the jet energy scale and resolution determined, and how is the spectrum unfolded? These questions are covered in Modules 7 and 9.

Pitfalls.
Comparing jet results across experiments with different R, constituent definitions or η ranges. Interpreting collimation of surviving jets as decoherence without considering quark/gluon selection bias.

6 Heavy flavour and quarkonia

Goal. Explain why charm, bottom and their bound states probe the QGP differently from light partons, and what the Heavy Flavor TG (HFTG) needs from the detector and software [S06].

6.1 Why heavy quarks

Charm ($m_c \approx 1.3$ GeV) and bottom ($m_b \approx 4.2$ GeV) quarks are much heavier than T and $\Lambda_{\rm QCD}$. Three consequences follow.

1. They are produced almost only in initial hard scatterings (thermal production is negligible at RHIC temperatures) and their number is conserved during the evolution. They therefore carry a memory of the whole history.
2. Their energy loss differs. Radiation from a heavy quark is suppressed inside a cone of angle $\theta_0 \simeq m_Q/E$ (the *dead-cone effect*); a 10 GeV bottom quark has $\theta_0 \approx 0.4$. Collisional loss is relatively more important. A mass hierarchy $\Delta E_g > \Delta E_c > \Delta E_b$ is expected.
3. At low $p_{\rm T}$ a heavy quark performs Brownian motion in the medium and partly thermalizes; its spatial diffusion coefficient D_s (often quoted as $2\pi T D_s$) is a transport property of the QGP. Charm flow (v_2 of D mesons) measures how strongly charm couples to the medium.

Hadronization. In the QGP a charm quark can combine with a nearby light quark (coalescence), which enhances baryons relative to mesons. Baryon/meson ratios (Λ_c/D^0) and vector/scalar ratios are therefore probes of hadronization [S06].

6.2 Quarkonia: a QGP thermometer

In the QGP, colour screening weakens the binding of $c\bar{c}$ and $b\bar{b}$ states once the Debye screening length becomes smaller than the state's radius. Larger, more weakly bound states dissociate at lower temperature (*sequential suppression*). For bottomonium the $\Upsilon(1S)$ is the most tightly bound and smallest, the $\Upsilon(2S)$ and $\Upsilon(3S)$ are larger. Their masses are 9.460, 10.023 and 10.355 GeV, so the $2S$–$1S$ splitting is only 563 MeV: separating the states in $\mu^+\mu^-$ or e^+e^- requires excellent momentum resolution, which drove the sPHENIX tracking design [5]. The key observable is the modification of the ratio $\Upsilon(2S)/\Upsilon(1S)$ in Au+Au relative to p+p [S06]. At RHIC, $b\bar{b}$ recombination is small, so the suppression pattern is cleaner than at the LHC. In sPHENIX the $\Upsilon \to e^+e^-$ channel uses the triggered data, track–calorimeter matching (electron identification from E/p in the EMCal) and high-p_{T} momentum resolution; it is statistically and technically harder than open heavy flavour [S06].

6.3 The HFTG programme

- **Strangeness**: K^0_S, Λ, Ξ, ϕ in p+p, Au+Au and O+O. High signal/background and well known physics make them benchmarks for tracking; the current focus includes strange-hadron ratios such as Λ/K^0_S [S06].
- **Open heavy-flavour hadrons**: $D^0 \to K^-\pi^+$, Λ_c, B mesons: ratios, R_{AA} and v_2. This depends strongly on momentum and pointing (DCA) resolution from the MVTX [S06].
- **Quarkonia**: $\Upsilon(1S, 2S, 3S)$ [S06].
- **Heavy-flavour jets and substructure**: explicit tagging (D^0 inside the jet cone) or implicit tagging (secondary-vertex displacement and impact-parameter tags combined with deep-learning algorithms); both need particle-flow style matching between trackers and calorimeters [S06]. Heavy-flavour energy correlators and fragmentation, and the transverse single-spin asymmetry of D mesons (Section 7), are also on the list.

6.4 How it is done in sPHENIX

Secondary-vertex reconstruction uses `KFParticle` (coresoftware package `KFParticle_sPHENIX`). KFParticle is an external package integrated into the sPHENIX pipeline; updates to it can change reconstruction in ways that are hard to distinguish from calibration changes, so the HFTG wants a KFParticle QA output in simulation as well as in data [S06]. High-priority near-term tasks named at the school were [S06]: dead-channel maps in simulation (especially TPC front-end electronics that are off), so that data/MC acceptance agrees; minimum-bias pion spectra with charge-dependent efficiency data vs simulation (a basic tracking test); KFParticle QA in simulation; integrating Ξ (cascade) and ϕ reconstruction into the TG's light-flavour framework; and yields versus trigger type and versus streaming readout.

Checks.

A heavy-flavour signal must appear at the right mass with a width consistent with simulation; its yield must scale sensibly with the number of events; and K^0_S and Λ yields for positive and negative daughters must be consistent once efficiencies are applied.

Pitfalls.

Data/MC acceptance mismatches from dead areas that exist in data but not in simulation; time dependence of dead maps within a run; treating a change in KFParticle version as a physics effect.

7 Cold QCD and spin physics with polarized protons

Goal. Understand the cold-QCD questions sPHENIX addresses with transversely polarized p+p collisions, the meaning of a transverse single-spin asymmetry, and the planned channels [S04,S44,S50].

7.1 The structure of the proton

In the collinear parton model a fast proton is described by parton distributions $f(x, \mu^2)$. At leading twist there are three collinear PDFs for quarks: the unpolarized f_1, the helicity distribution g_1 (quark spin along the proton spin for a longitudinally polarized proton) and the transversity h_1 (quark transverse spin in a transversely polarized proton). The proton spin decomposes as $\frac{1}{2} = \frac{1}{2}\Delta\Sigma + \Delta G + L_q + L_g$: quark spins contribute only about a third, and the gluon helicity ΔG and orbital angular momenta L make up the rest. Orbital motion implies transverse momentum k_T of partons, which leads to *transverse-momentum-dependent* distributions (TMDs).

7.2 Transverse single-spin asymmetries

With a transversely polarized beam, particles can be produced preferentially to the left or right of the plane spanned by the beam and the spin. The *analyzing power* is

$$A_N = \frac{\sigma_L - \sigma_R}{\sigma_L + \sigma_R} \quad \text{or equivalently} \quad A_N = \frac{1}{P}\,\frac{\sigma^{\uparrow} - \sigma^{\downarrow}}{\sigma^{\uparrow} + \sigma^{\downarrow}} \text{ at fixed side,} \tag{19}$$

with beam polarization P. Large asymmetries (tens of percent at forward rapidity) have been observed from low energy up to the highest RHIC energies [S04]. Naive collinear pQCD predicts tiny asymmetries, proportional to the quark mass over the hard scale, so the large A_N requires new dynamics: correlations between spin and transverse momentum.

Mechanisms [S04]:

- **Sivers effect** (initial state): a correlation between the proton spin and the transverse momentum of an unpolarized parton inside it, $\vec{S} \cdot (\hat{p} \times \vec{k}_T)$.
- **Collins effect** (final state): a transversely polarized quark fragments into hadrons with a left–right asymmetry around its direction; it couples to transversity h_1.

Sivers and Collins functions are defined in the TMD framework, which needs two scales: a hard scale Q and a measured transverse momentum $p_{\mathrm{T}} \ll Q$ (for example, the imbalance of a dijet or γ+jet pair). Inclusive single-hadron production in p+p has only one hard scale (p_{T}). It is described instead in the *collinear twist-3* framework, where the asymmetry comes from multi-parton (quark–gluon or tri-gluon) correlation functions in the polarized proton, in the unpolarized proton (combined with transversity) or in fragmentation. The two frameworks are related through k_T-moments of the TMDs [S04]. The TMD lecture at the workfest [S50] and the cold-QCD lecture series [S44] (on RHIC polarized p+p running, on polarimetry and on TMDs) developed these ideas further.

7.3 Measuring A_N in practice

RHIC fills alternate the spin direction bunch by bunch, so both spin states are recorded at the same time with the same detector. With yields $N^{\uparrow}$, $N^{\downarrow}$ and the relative luminosity $\mathcal{R} = \mathcal{L}^{\uparrow}/\mathcal{L}^{\downarrow}$,

$$A_N = \frac{1}{P}\,\frac{N^{\uparrow} - \mathcal{R}N^{\downarrow}}{N^{\uparrow} + \mathcal{R}N^{\downarrow}}. \tag{20}$$

With a detector on both sides (left L, right R) the square-root formula cancels both relative luminosity and acceptance to first order:

$$\epsilon = \frac{\sqrt{N_L^{\uparrow} N_R^{\downarrow}} - \sqrt{N_L^{\downarrow} N_R^{\uparrow}}}{\sqrt{N_L^{\uparrow} N_R^{\downarrow}} + \sqrt{N_L^{\downarrow} N_R^{\uparrow}}}, \qquad A_N = \epsilon / P. \tag{21}$$

For a full-azimuth detector like sPHENIX one fits the ϕ modulation, $\epsilon(\phi) = P A_N \cos(\phi - \phi_{\rm spin})$. The beam polarization P (well below 100%) comes from the RHIC polarimeters (Module 2). For a resonance such as the π^0 the measured asymmetry in the mass window mixes signal and combinatorial background with fraction r:

$$A_N^{\rm sig} = \frac{A_N^{\rm meas} - r\, A_N^{\rm bkg}}{1 - r}, \tag{22}$$

where $A_N^{\rm bkg}$ is measured in side bands [S04].

7.4 The sPHENIX cold-QCD channels

sPHENIX recorded transversely polarized p+p collisions in 2024 and 2025: a large calorimeter-triggered sample, an all-detector sample and a tracker streaming sample [S04]. The channels are [S04]:

- **Neutral mesons** (π^0, η) at mid-rapidity, reconstructed as EMCal cluster pairs; both initial- and final-state effects contribute. Corrections for background fraction and background asymmetry are applied as above.
- **Inclusive jets** (anti-k_t, $R = 0.4$, several η ranges): probe the interplay between initial-state correlations and final-state jet production; cross-checked against STAR.
- **Direct photons**: the fragmentation contribution is strongly suppressed, so the asymmetry is mostly initial state; at mid-rapidity the $qg \to q\gamma$ Compton process dominates and the measurement constrains the tri-gluon correlator, since the quark contribution is expected to be small. The isolated-photon cross section [4] is the unpolarized baseline.
- **γ+jet** (the "golden channel"): measuring both photon and jet fixes the parton kinematics event by event. At leading order $x_{1,2} = \frac{p_{\rm T}}{\sqrt{s}} (e^{\pm\eta_\gamma} + e^{\pm\eta_{\rm jet}})$; for a 10 GeV photon and jet both at $\eta = 0$, $x_1 = x_2 = 0.1$. The recoil jet also rejects fake photons from π^0 and η decays, and azimuthal correlations help separate Compton scattering from annihilation [S04].
- D^0 A_N: charm production at RHIC is dominated by gluon–gluon fusion, so it probes tri-gluon correlations; this uses the streaming tracker data [S04,S01].
- **Dijets**: the spin-dependent k_T imbalance (Sivers) with jet-charge tagging to enrich u or d jets.
- **Hadron-in-jet** (Collins) and **dihadron-in-jet** (interference fragmentation functions coupling to transversity);
- further channels: $\Lambda\bar{\Lambda}$ spin correlations, Λ spin transfer, ϕ, K_S^0 and Λ A_N [S04].

Connect to your analysis.

The same EMCal π^0 and η yields you extract for the cross section are the input to the cold-QCD A_N analysis, split by bunch spin state. If you ever need a spin-sorted yield, the bunch number from GL1 and the spin pattern from the spin database (`SpinDBContent` in package `uspin`; Module 2) are what you need. Your signal/background fit per $p_{\rm T}$ bin directly provides the background fraction r.

8 From PHENIX to sPHENIX: a short history

Goal. Know how sPHENIX came to be, because many design choices (and quirks) make sense only in that light [S02,S08].

8.1 RHIC, RIKEN and PHENIX

RHIC was built in the 1990s in a tunnel originally dug for a proton collider; first collisions came in 2000, with the emphasis on discovering the QGP [S02]. From the start the Japanese institute RIKEN, at the initiative of T. D. Lee, supported the polarized-proton programme: the RIKEN BNL Research Center (RBRC) began in 1998, provided accelerator components needed for polarized beams, and worked with PHENIX on the spin programme; its fellowship system trained a generation of theorists and experimentalists now in permanent positions in the US and Japan [S02]. PHENIX ran from 2000 to 2016 and made, among many results, the first observation of high-p_{T} hadron suppression [9, 10].

8.2 The road to sPHENIX

- **2009**: the BNL associate director asked STAR and PHENIX for decadal plans: compelling science needing upgrades, why RHIC is the right place, and plans for a future electron–ion collider [S02].
- **2012**: the first sPHENIX proposal was submitted to BNL [S08]. Decommissioned solenoids from other experiments were examined and found unsuitable. The original HCal concept already had tilted steel plates that double as the magnet flux return, with scintillating tiles read out by wavelength-shifting fibres [S02]. Reviewers found the science case strong but asked for tracking upgrades, questioned the novel EMCal design, and asked for large contingency on the magnet [S02].
- **2013**: DOE approved the transfer of the BaBar solenoid from SLAC to BNL; it arrived in 2015 [S02]. The calorimeters were redesigned around it; the EMCal became tungsten powder with scintillating fibres [S02].
- **2014–2015**: further reviews; debate in the community; the 2015 Long Range Plan endorsed sPHENIX [S01,S02]. The sPHENIX proposal was published [5].
- **December 2015**: the collaboration was formed at a meeting of about 70 participants (not all from PHENIX) at Rutgers; G. Roland and D. Morrison were elected co-spokespersons [S02].
- **2016–2019**: DOE project milestones CD-0 (September 2016), CD-1/3A (August 2018) and PD-2/3 (September 2019) [S08]. Cost pressure forced descoping, followed by scope recovery: Chinese institutions (Fudan, Peking University, CIAE) took on the high-η EMCal sectors, RIKEN contributed the INTT, BNL capital projects provided the MVTX, inner HCal and TPC outer tracker, and NSF funding through Lehigh provided the sEPD [S02].
- **Construction**: TPC at several US universities; EMCal at UIUC and the Chinese institutions; outer HCal at GSU and BNL; inner HCal at ISU; INTT at BNL, RIKEN and Purdue; sEPD at Lehigh and Colorado; TPOT at Saclay and BNL; MVTX at CERN, LBNL, MIT and LANL; the MBD at BNL [S02]. The ZDC and MBD (the PHENIX beam–beam counter) were reused from PHENIX [S08].
- **2023**: after an operations-readiness review focused on safety, Run 2023 began in May 2023 [S02,S08].
- **2026**: the last RHIC run ended on 6 February 2026 [S08]. The RHIC tunnel and injectors now serve the Electron–Ion Collider (EIC) project.

8.3 The collaboration today

sPHENIX recorded by far the largest data volume of any RHIC experiment [S01]. Foundational work on calibration and reconstruction continues, both to enable the physics and to demonstrate the "Ultimate Performance Parameters" (UPPs), the performance goals of the project [S01]. Many PPGs exist; the first papers are published or submitted, and the school/workfest pushed towards the next wave of results for Quark Matter 2027 [S01]. New members report on their work after 6–12 months for authorship review [S01].

9 Where neutral mesons fit

Goal. Pull the threads together: see every way your π^0 and η measurements feed the whole programme.

Measurement	Physics role	TG
p+p π^0, η cross section	pQCD test (NLO with fragmentation functions); constrains gluon→hadron fragmentation; reference for R_{AA}; EMCal energy-scale validation	JSTG
η/π^0 ratio	approximately constant at high p_{T} (m_T scaling at low p_{T}); a cross-check of fragmentation and of the photon background	JSTG
Au+Au π^0 R_{AA}, R_{CP}	parton energy loss with high statistics and particle identification	JSTG
π^0 v_2	bulk flow at low p_{T}; path-length dependence of energy loss at high p_{T}	Bulk
π^0, η A_N	twist-3 initial- and final-state correlators	Cold QCD
π^0/η decay photons	dominant background for direct/isolated photons and γ+jet; must be modelled or tagged	JSTG, Cold QCD
π^0 mass peak	EMCal tower-by-tower calibration (Modules 3, 7)	Calibration

Table 1: Roles of neutral-meson measurements in the sPHENIX programme.

Connect to your analysis.
(1) Your cross section is simultaneously a physics result and a calibration closure test: if the EMCal energy scale is off by 1%, a spectrum falling as p_{T}^{-8} shifts by about 7–8% at fixed p_{T}. (2) The η/π^0 ratio from the same data is a free internal consistency check. (3) Your Au+Au R_{CP} needs only Glauber N_{coll} ratios; once your p+p result exists, R_{AA} follows with the T_{AA} normalization uncertainty instead. (4) Decay photons from your π^0s and ηs are the background in the isolated-photon analyses; your η/π^0 ratio can be useful to those PPGs.

10 Hands-on exercises on SDCC (optional)

These are suggestions; they were not run while writing this text, and the text does not depend on them.

1. **Kinematics.** In ROOT, write a macro that converts $\eta \leftrightarrow \theta$ and $y \leftrightarrow \eta$ for a π^0 of $p_{\mathrm{T}} = 2$ GeV and for a proton of $p_{\mathrm{T}} = 0.5$ GeV; check the numbers of Section 2.
2. **Toy R_{AA}.** Generate a power-law spectrum ($n = 8$), shift each entry by a constant fraction ϵ and by a fluctuating fraction (e.g. ϵ from an exponential distribution with the same mean). Compare the two R_{AA} curves; this shows why R_{AA} does not fix $P(\epsilon)$.
3. **Toy flow.** Generate events with v_2 drawn from a Gaussian ($\bar{v} = 0.06$, $\sigma = 0.02$), 500 particles per event; compute $v_2\{2\}$ and $v_2\{4\}$ from Q-vectors and compare with the formulas of Section 3.

4. **PYTHIA8.** On SDCC (`source /opt/sphenix/core/bin/sphenix_setup.sh -n`), run a few thousand PYTHIA8 hard-QCD events at 200 GeV with $\hat{p}_T > 5$ GeV and histogram the fraction of π^0s above 5 GeV that come from gluon versus quark fragmentation (use the mother history). Check the current PYTHIA8 version and setup in the sPHENIX environment first.
5. **Checkpoint (syllabus).** Read the two first Au+Au papers [1, 2] and write a one-page map of which sPHENIX measurement constrains which QGP property (energy density, η/s, $\hat{q}$, D_s, screening, hadronization).

11 Self-test questions

Q1. Using Eq. (1), estimate α_s at $Q = 2$ GeV ($n_f = 4$) and at $Q = 50$ GeV ($n_f = 5$). What does the difference imply for jets versus the bulk?

Q2. Why is pseudorapidity, not rapidity, the natural variable for detector acceptance? When does the difference matter?

Q3. Derive Eq. (6). Which input is best known from experiment and which is model dependent?

Q4. Explain why v_2 is largest in mid-central collisions.

Q5. If v_2 has mean 0.06 and standard deviation 0.02, estimate $v_2\{2\}$ and $v_2\{4\}$.

Q6. A spectrum falls as $E\,d^3N/dp^3 \propto p_{\mathrm{T}}^{-7}$. What R_{AA} results from a constant 15% momentum loss? What if the slope were 5?

Q7. Derive $t_f \simeq \sqrt{\omega/\hat{q}}$ and $\omega_c = \hat{q}L^2$, and explain the L^2 dependence of radiative energy loss.

Q8. What is colour coherence and which angle separates coherent from decoherent splittings?

Q9. Give two reasons why surviving jets in Au+Au might appear more collimated than in p+p.

Q10. Why is the $\Upsilon(2S)/\Upsilon(1S)$ ratio a better observable than the $\Upsilon(1S)$ R_{AA} alone, and why does it need good momentum resolution?

Q11. What is the dead-cone angle of a 5 GeV charm quark, and what does it imply?

Q12. Why does inclusive π^0 A_N in p+p require the twist-3 framework rather than TMDs?

Q13. Write the background-corrected A_N for a π^0 peak with 30% background, $A_N^{\mathrm{meas}} = 0.010$ and $A_N^{\mathrm{bkg}} = 0.004$.

Q14. Name the four topical groups and one flagship measurement each.

Q15. List four ways in which the p+p π^0 cross section matters beyond being a cross section.

Glossary

A_N transverse single-spin asymmetry (analyzing power).

Anti-k_t
the standard jet clustering algorithm; produces cone-like jets of radius R.

Colour coherence
a jet's prongs radiate as one charge if the medium cannot resolve them ($\theta < \theta_c$).

Cumulant $v_n\{k\}$
flow from k-particle correlations with lower-order correlations removed.

Dead cone
suppression of gluon radiation from a heavy quark at angles $\lesssim m/E$.

Eccentricity ε_n
shape coefficient of the initial energy density.

$\hat{q}$ jet transport coefficient: transverse momentum squared per unit path length.

Hydrodynamization
onset of hydrodynamic behaviour before full equilibrium.

LPM effect
destructive interference that suppresses radiation during long formation times.

N_{part}, N_{coll}
Glauber numbers of participating nucleons and binary nucleon–nucleon collisions.

PPG paper preparation group; the unit that produces one paper.

R_{AA}, R_{CP}
nuclear modification factor; central-to-peripheral ratio.

Sivers, Collins
initial-state and final-state spin–momentum correlations (TMDs).

TG topical group: Bulk, Jet Structure, Heavy Flavor, Cold QCD.

Twist-3
collinear framework with multi-parton correlators, used when only one hard scale is measured.

UPP Ultimate Performance Parameters, the sPHENIX project performance goals.

x_T $2p_{\mathrm{T}}/\sqrt{s}$, the scaling variable of hard processes.

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A Answers

A1. At 2 GeV: $12\pi/(25\ln 100) = 37.7/115 = 0.33$. At 50 GeV: $12\pi/(23\ln 62500) = 37.7/254 = 0.15$. Jets (large Q) are in the perturbative regime; the bulk ($Q \sim T \sim 0.3$ GeV) is not, so it needs hydrodynamics and lattice QCD.

A2. A detector measures angles, and η depends only on θ; y also needs the mass. They differ when p_{T} is not large compared with m: for a 0.5 GeV proton at $\theta = 90°$, $\eta = 0 = y$, but at $\eta = 1$ the proton has $y \approx 0.5$.

A3. See Section 3: slice thickness $dz = \tau\, dy$, volume $A_T \tau\, dy$. dE_T/dy is measured; A_T comes from Glauber geometry; τ is model dependent, hence $\varepsilon\tau$ is quoted.

A4. ε_2 grows towards peripheral collisions, but the system becomes smaller and shorter lived so the hydrodynamic response weakens; the product peaks at mid-centrality.

A5. $v_2\{2\} = \sqrt{0.0036 + 0.0004} = 0.063$; $v_2\{4\} \approx \sqrt{0.0036 - 0.0004} = 0.057$.

A6. $R_{AA} = (1-\epsilon)^{n-2}$: $0.85^5 = 0.44$ for $n = 7$; $0.85^3 = 0.61$ for $n = 5$. Steeper spectra amplify suppression.

A7. $t_f = \omega/k_\perp^2$ and $k_\perp^2 = \hat{q} t_f$ give $t_f^2 = \omega/\hat{q}$. Requiring $t_f < L$ gives $\omega < \hat{q}L^2$. The spectrum $\omega\, dI/d\omega \propto \sqrt{\omega_c/\omega}$ integrates to $\Delta E \propto \omega_c = \hat{q}L^2$.

A8. Radiation from a pair of colour charges at small opening angle interferes as if from their total charge. The medium resolves the pair only if $\theta L > (\hat{q}L)^{-1/2}$, i.e. $\theta > \theta_c = 1/\sqrt{\hat{q}L^3}$.

A9. (a) Wide jets (resolved, decoherent, more prongs) lose more energy and migrate to lower p_{T}, so the surviving sample at fixed p_{T} is narrower. (b) Gluon jets (wider) lose more energy than quark jets, enriching the sample with narrow quark jets.

A10. The ratio cancels many systematic uncertainties (luminosity, $N_{\rm coll}$, efficiencies largely) and directly tests sequential melting. The $2S$ and $1S$ peaks are only 563 MeV apart at 10 GeV, so the mass resolution must be well below 100 MeV.

A11. $\theta_0 \approx 1.3/5 \approx 0.26$ rad. Radiation inside that angle is suppressed, so charm loses less energy radiatively than light quarks of the same energy at low $p_{\rm T}$.

A12. Only one hard scale ($p_{\rm T}$) is measured. TMD factorization needs two scales (a hard Q and a small measured transverse momentum); with one scale, collinear twist-3 correlators describe the asymmetry.

A13. $A_N^{\rm sig} = (0.010 - 0.3 \times 0.004)/0.7 = 0.0088/0.7 = 0.0126$.

A14. Bulk: $dE_T/d\eta$, $dN_{ch}/d\eta$, flow. Jet Structure: jet R_{AA}, dijets, substructure, photons. Heavy Flavor: Υ states, D mesons, HF jets. Cold QCD: transverse single-spin asymmetries (π^0, jets, photons, D^0).

A15. pQCD and fragmentation-function test; reference for R_{AA}; EMCal energy-scale validation; spin-asymmetry input; decay-photon background estimate for direct photons; η/π^0 consistency check.